

COASTAL CAROLINA, NC, USA

WMO: 723095

Lat: 35.068N

Lon: 77.048W

Elev: 6

StdP: 101.26

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93719

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.3	-3.3	-14.5	1.1	-2.1	-11.9	1.3	0.4	10.2	11.8	9.0	8.2	2.4	350	0.407

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	34.1	25.6	32.9	25.1	31.8	24.7	26.8	31.7	26.2	30.8	25.7	29.9	3.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.6	20.8	28.9	25.0	20.1	28.4	24.5	19.5	27.8	83.8	31.8	81.1	30.7	78.7	30.0	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.5	7.6	6.8		-8.6	36.3	2.0	1.2	-10.0	37.2	-11.1	37.9	-12.2	38.5	-13.7	39.4
			DB	-9.5	28.3	1.9	0.8	-10.8	28.9	-11.9	29.3	-13.0	29.8	-14.3	30.4
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.3	6.9	8.6	11.8	16.9	21.1	25.3	26.8	26.2	23.6	18.2	12.5	9.1
	DBStd	8.18	5.96	5.13	5.04	4.36	3.60	2.51	2.00	1.97	2.67	4.27	4.58	5.45
	HDD10.0	370	134	81	39	3	0	0	0	0	2	26	85	
	HDD18.3	1465	357	276	209	78	16	0	0	0	2	56	181	291
	CDD10.0	3031	37	41	97	209	344	459	521	503	407	254	101	57
	CDD18.3	1084	1	2	7	35	102	209	262	245	158	51	7	3
	CDH23.3	9047	4	14	58	271	806	1838	2483	2135	1071	324	33	12
	CDH26.7	3220	0	1	6	57	251	721	1001	810	311	62	2	0
(14) Wind	WSAvg	2.9	3.1	3.3	3.6	3.6	3.1	2.8	2.6	2.4	2.7	2.6	2.7	2.8
(15) Precipitation	PrecAvg	1386	111	99	107	84	115	121	179	168	133	79	78	92
	PrecMax	1846	205	238	233	212	256	220	353	331	354	249	195	248
	PrecMin	1056	30	32	15	18	25	50	74	43	22	3	7	8
	PrecStd	197	50	48	48	45	50	48	72	81	91	57	45	56
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.6	25.5	27.6	30.5	33.4	35.4	35.4	35.1	32.6	30.5	26.3	24.9
		MCWB	18.2	18.6	18.4	20.6	23.1	25.1	26.4	26.0	24.4	23.1	20.9	19.9
	2%	DB	21.3	22.3	25.0	28.3	31.2	33.5	34.0	33.4	31.2	28.4	24.1	22.5
		MCWB	17.0	16.5	17.7	19.8	22.7	24.8	26.1	25.6	24.0	22.0	18.7	18.8
	5%	DB	19.2	19.9	22.7	26.6	29.4	32.1	32.8	32.1	29.9	26.7	22.4	20.6
		MCWB	15.7	15.6	16.4	19.1	21.9	24.4	25.6	25.4	23.3	21.0	18.3	17.8
	10%	DB	16.9	17.8	20.6	24.6	27.7	30.7	31.6	30.8	28.5	25.0	20.6	18.6
		MCWB	13.9	14.0	15.8	18.0	21.0	23.8	25.1	24.6	23.0	20.3	17.2	16.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.6	19.9	20.4	22.9	24.8	26.8	27.9	27.7	26.2	25.0	22.4	21.4
		MCDB	21.8	23.5	23.9	26.9	30.5	32.4	33.1	32.5	30.1	27.6	24.6	23.5
	2%	WB	18.2	18.5	19.2	21.5	23.8	25.8	26.9	26.8	25.4	23.8	20.7	19.7
		MCDB	20.6	21.3	22.7	25.8	28.8	31.1	32.1	31.2	28.9	26.5	22.5	21.7
	5%	WB	16.4	16.7	18.0	20.4	22.9	25.3	26.3	26.1	24.8	22.8	19.2	18.2
		MCDB	18.9	19.0	21.3	24.5	27.4	30.4	31.1	30.3	28.0	25.3	21.4	20.0
	10%	WB	14.3	14.5	16.7	19.3	22.1	24.7	25.7	25.6	24.2	21.6	17.7	16.6
		MCDB	16.3	17.4	19.9	23.2	26.1	29.1	30.1	29.3	27.0	24.0	20.0	18.6
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	11.8	12.0	11.9	10.9	10.0	9.3	9.0	8.9	10.7	11.7	10.9
		MCDBR	13.9	14.5	14.5	13.7	12.4	11.4	10.6	10.5	10.3	11.4	13.0	12.4
		MCWBR	10.1	9.7	8.1	6.8	5.4	4.5	4.2	4.2	4.3	5.8	8.4	9.0
	5% WB	MCDBR	12.5	13.3	11.8	11.5	10.5	10.2	9.7	9.4	8.6	8.7	10.8	11.1
		MCWBR	10.4	10.3	8.2	6.7	4.9	4.4	4.3	4.2	4.0	5.1	8.3	9.2
(40) Clear-Sky Solar Irradiance	taub		0.317	0.330	0.357	0.399	0.444	0.493	0.522	0.507	0.420	0.369	0.337	0.324
	taud		2.534	2.492	2.437	2.320	2.234	2.142	2.077	2.112	2.357	2.479	2.519	2.544
	Ebn at Noon		891	916	915	887	847	802	776	780	839	860	858	859
	Edh at Noon		81	94	108	127	140	154	163	155	116	94	82	76
(44) All-Sky Solar Radiation	RadAvg		2.60	3.32	4.38	5.54	6.10	6.36	6.03	5.47	4.54	3.80	2.88	2.31
	RadStd		0.16	0.33	0.31	0.37	0.43	0.34	0.35	0.32	0.31	0.42	0.21	0.18

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+197	+131
(47) Regional (45 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+161	+115

Nomenclature: See separate page